

Review

Machine Learning-Based Intrusion Detection Methods in IoT Systems: A Comprehensive Review

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Abstract: The rise of the Internet of Things (IoT) has transformed our daily lives by connecting objects to the Internet, thereby creating interactive, automated environments. However, this rapid expansion raises major security concerns, particularly regarding intrusion detection. Traditional intrusion detection systems (IDSs) are often ill-suited to the dynamic and varied networks characteristic of the IoT. Machine learning is emerging as a promising solution to these challenges, offering the intelligence and flexibility needed to counter complex and evolving threats. This comprehensive review explores different machine learning approaches for intrusion detection in IoT systems, covering supervised, unsupervised, and deep learning methods, as well as hybrid models. It assesses their effectiveness, limitations, and practical applications, highlighting the potential of machine learning to enhance the security of IoT systems. In addition, the study examines current industry issues and trends, highlighting the importance of ongoing research to keep pace with the rapidly evolving IoT security ecosystem.

Keywords: IoT; security; intrusion detection; machine learning; DoS



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1. Introduction

In the ever-evolving field of information technology, the Internet of Things (IoT) has emerged as a groundbreaking innovation, integrating everyday objects into the Internet, making our environment more interactive and automated [1]. However, with the exponential growth of IoT, significant security challenges have also emerged, attracting the attention of researchers and industry professionals [2].

Intrusion detection systems (IDSs) play a crucial role in protecting computer systems by detecting and responding to malicious activities. They continuously monitor networks to identify abnormal behaviors or potential attacks, thereby maintaining the integrity, confidentiality, and availability of systems. Traditional IDS methods, designed for standard networks, face major challenges in diverse IoT networks [3]. Their lack of flexibility regarding the variety of IoT devices and protocols reduces their effectiveness. Advanced attacks, including zero-day attacks, easily surpass these IDSs, which rely on known signatures. Additionally, their inability to process large volumes of data leads to performance and scalability issues. Their dependence on manual signature updates increases the vulnerability of IoT systems. Traditional IDSs also struggle to differentiate legitimate behaviors from malicious ones in varied IoT traffic patterns, thus increasing the risk of errors.

In response to these limitations, machine learning has emerged as a promising solution capable of adapting and responding to complex and evolving threats in the IoT environment [4]. Machine learning-based IDSs can learn from historical data to detect abnormal behaviors, offering an enhanced ability to identify new and unknown attacks. Furthermore, these systems can process large volumes of data and continuously improve through machine learning.

This review delves into various machine learning approaches, including supervised, unsupervised, and deep learning methods, evaluating their effectiveness in intrusion detection. The discussion also covers practical applications and industry implications, highlighting the importance of ongoing research to enhance IoT system security. By providing a comparative analysis of existing methods, this review aims to identify the best practices and current gaps to guide future research.

The core of this review focuses on intrusion detection systems in IoT, exploring the basic principles, significance, and traditional detection methods. It then analyzes the limitations of these traditional approaches and presents machine learning as a possible solution. The subsections detail supervised, unsupervised, and deep learning, evaluating their effectiveness and limitations in the context of IoT systems. Finally, the review concludes with a discussion on emerging trends and challenges in intrusion detection for IoT and offers a critical analysis of current methods as well as suggestions for future research.

2. Materials and Methods

A systematic review is an explicit and reproducible research methodology that identifies all relevant studies and summarizes the state of the art, in order to answer one or more fundamental research questions on a particular topic. In this section, we will present the detailed methodology used to conduct our systematic review of machine learning-based intrusion detection methods in IoT systems. Our approach follows the guidelines described in the PRISMA (preferred reporting items for systematic reviews and meta-analyses) method.

2.1. Eligibility Criteria

To ensure that only the most relevant studies were included in this systematic review, we defined specific inclusion and exclusion criteria. These criteria were systematically applied when evaluating each article identified in our initial search. The Table 1 below summarizes these criteria:

Table 1. Eligibility criteria for the systematic review.

Criterion	Description
Language	Only articles written in English.
Period	Only articles published in the last 10 years (between January 2014 and May 2024) due to the rapid developments in the field of IoT security.
Main topic	Only articles with the main topic of intrusion detection in IoT systems.
Techniques	Only articles addressing the topic with machine learning (ML)-based techniques.
Evaluation	Only peer-reviewed articles published in recognized scientific journals.

Scientific interest in the use of machine learning has grown over the past decade. The articles included in this review were published between 2013 and 2024. The following two graphs show the evolution of research over the last few years. Figure 1 shows the distribution of literature and systematic reviews on the subject over the last 5 years, while Figure 2 shows the evolution of publications relevant to our study over the last 10 years.

2.2. Data Sources and Search Strategy

Data Sources

For the identification and collection of articles related to our research, a literature search was performed across several electronic databases, including IEEE Xplore, PubMed, Scopus, and Google Scholar. These databases were selected for their relevance to the topic and scope of the search. Fields considered in search queries included the title, abstract, and keywords. Searches were formulated using several keywords corresponding to the eligibility criteria and using Boolean operators (AND, OR, NOT) to efficiently query the scientific publication databases. In addition, additional articles were identified from the bibliographies of the included articles. The Table 2 summarizes the queries launched in the various databases cited.

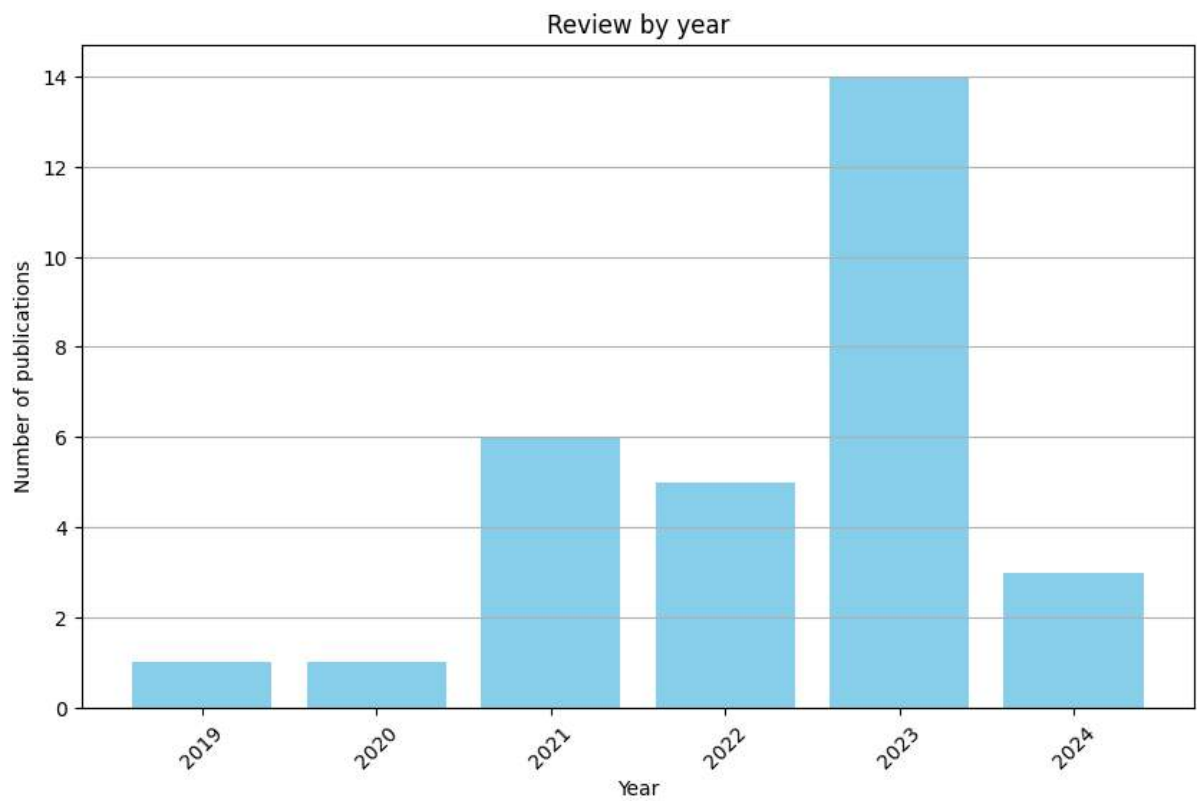


Figure 1. Distribution of reviews over the past 5 years (PubMed Database).

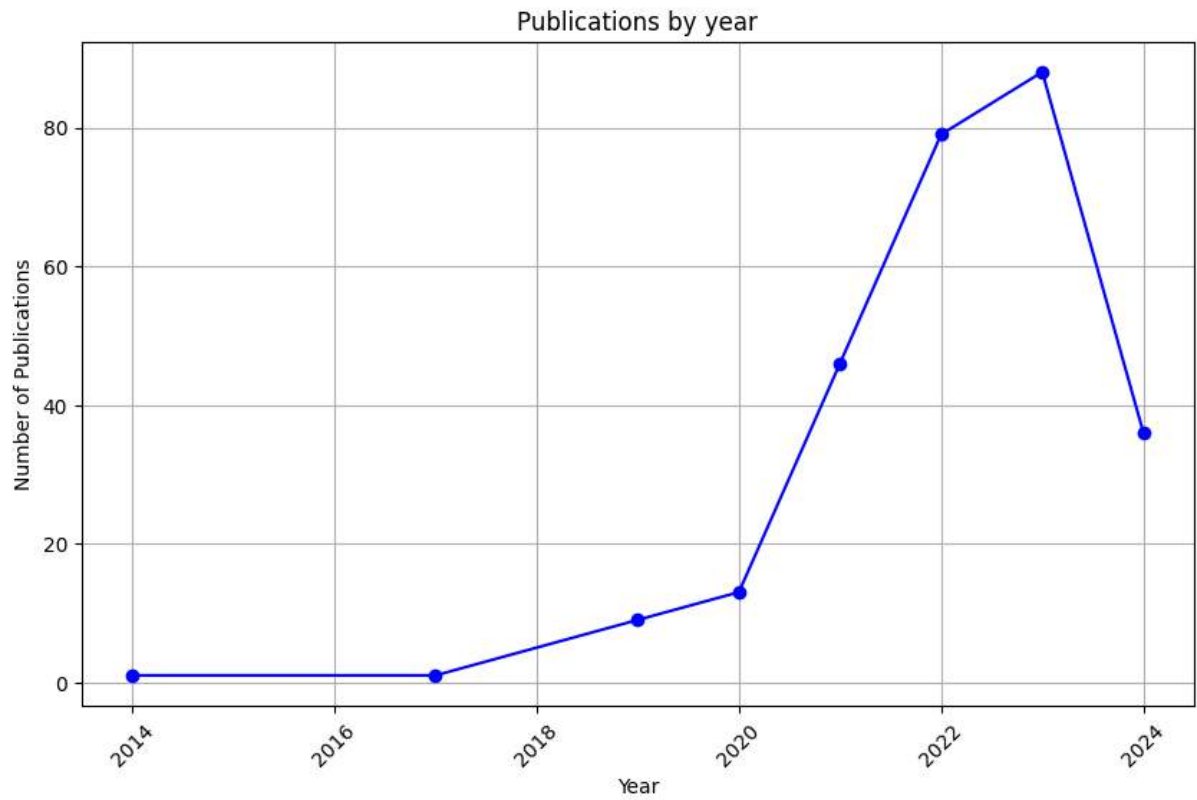


Figure 2. Distribution of documents by year (PubMed database).

Table 2. Search queries for databases.

Database	Search Query
IEEE Xplore	("IoT" OR "Internet of Things") AND ("intrusion detection" OR "anomaly detection" OR "cybersecurity") AND ("machine learning" OR "artificial intelligence" OR "deep learning" OR "KNN" OR "SVM" OR "GAN" OR "ANN" OR "logistic regression" OR "Random Forest") AND (LIMIT-TO (PUBYEAR, 2014–2024)) AND (LIMIT-TO (LANGUAGE, "English"))
PubMed	((("IoT" OR "Internet of Things") AND ("intrusion detection" OR "anomaly detection" OR "cybersecurity") AND ("machine learning" OR "artificial intelligence" OR "deep learning" OR "KNN" OR "SVM" OR "GAN" OR "ANN" OR "logistic regression" OR "Random Forest") AND ("2014/01/01"[PDAT]: "2024/05/31"[PDAT]) AND English[lang])
Scopus	TITLE-ABS-KEY ((("IoT" OR "Internet of Things") AND ("intrusion detection" OR "anomaly detection" OR "cybersecurity") AND ("machine learning" OR "artificial intelligence" OR "deep learning" OR "KNN" OR "SVM" OR "GAN" OR "ANN" OR "logistic regression" OR "Random Forest")) AND NOT (DOCTYPE ("re")) AND PUBYEAR > 2013 AND PUBYEAR < 2025 AND (LIMIT-TO (LANGUAGE, "English"))
Google Scholar	("IoT" AND "intrusion detection" AND "cybersecurity") AND ("machine learning" OR "artificial intelligence" OR "deep learning" OR "KNN" OR "SVM" OR "GAN" OR "ANN" OR "logistic regression" OR "Random Forest") AND (LIMIT-TO (PUBYEAR, 2014–2024)) AND (LIMIT-TO (LANGUAGE, "English"))

2.3. Search Strategy

We searched the IEEE Xplore, PubMed, Scopus, and Google Scholar databases using specific search terms to capture relevant articles. Fields considered in the search queries included the title, abstract, and keywords. Searches were formulated using several keywords corresponding to the eligibility criteria and using Boolean operators (AND, OR, NOT) to efficiently query scientific publication databases. The search terms used for each category were as follows:

- IoT: "IoT", "Internet of Things", "IoT system".
- Intrusion detection: "intrusion detection", "anomaly detection", "cybersecurity".
- Machine learning: "machine learning", "artificial intelligence", "ML", "AI", "deep learning", "supervised learning", "unsupervised learning", "neural network", "random forest", "support vector machine", "SVM", "Random Forest", "Decision Tree", "DNN", "ANN", "KNN", "GAN", "logistic regression", "ANN".
- Challenges: "security challenges", "IoT security issues", "cybersecurity challenges", "AI challenges", "threat detection challenges", "IoT vulnerabilities", "AI limitations in IoT security".

2.4. Study Selection

We used a reference management tool to record the references, eliminate duplicates, and create a unique database of references after querying the databases. To select articles from the initial database, we applied a three-step process.

- Title evaluation;
- Abstract and keyword evaluation;
- Full-text evaluation.

The aim was to remove irrelevant studies during steps (1) and (2), and then review the remaining documents using the eligibility criteria specified in step (3). Finally, during the eligibility phase, we compiled the included studies into our final database and noted the main reasons for excluding other articles based on the given criteria.

The entire flowchart of the selection process, including identification, screening, eligibility, and inclusion, is shown in Figure 3.

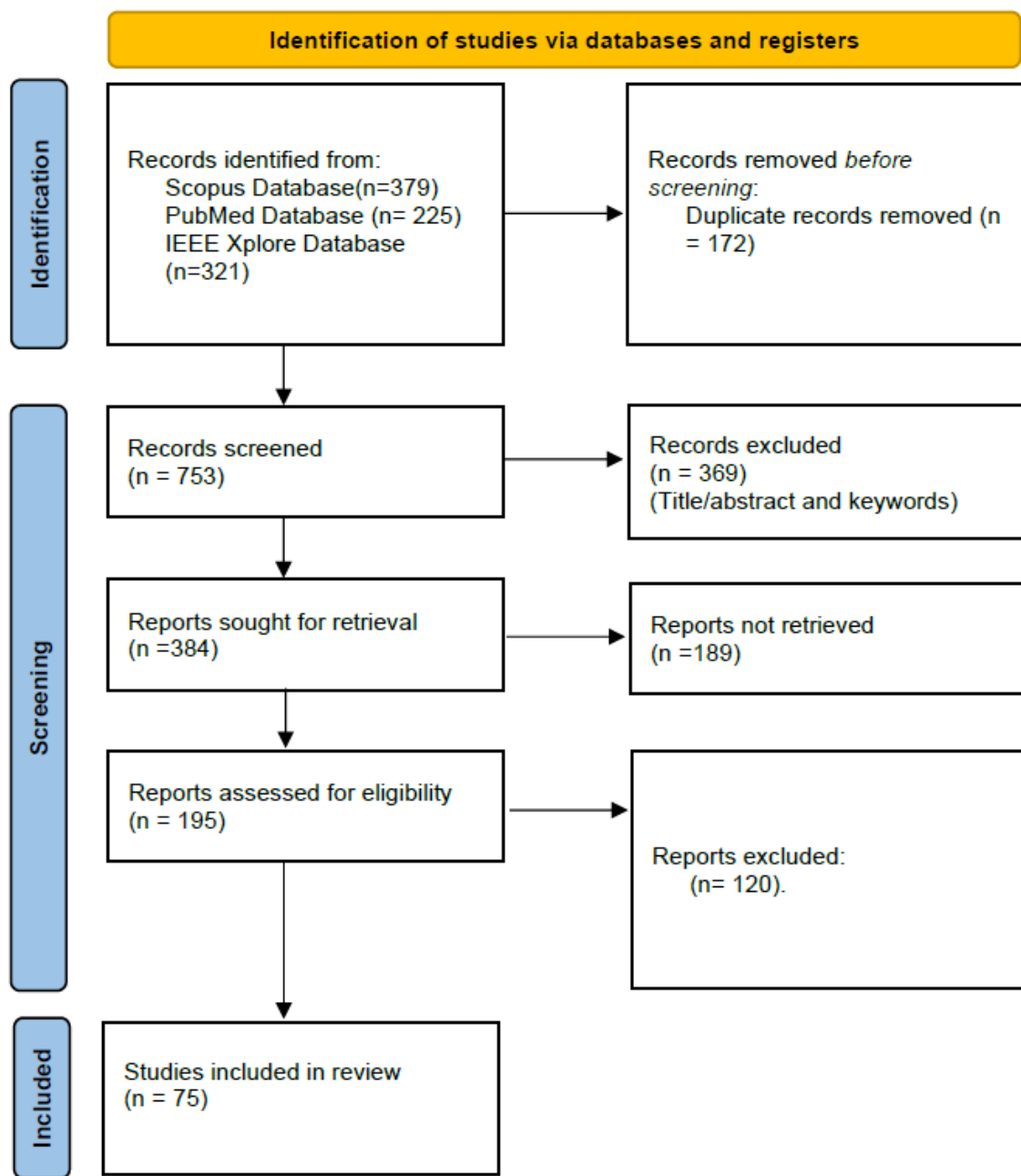


Figure 3. The PRISMA flow diagram.

3. Internet of Things

3.1. Definition and Growth of IoT

The Internet of Things (IoT) is a concept that has gained momentum in the early 21st century. According to Ashton [1], who is credited with introducing the term, IoT refers to a network of physical objects equipped with sensors, software, and other technologies, enabling them to connect and exchange data with other devices and systems over the Internet. This definition highlights the transformation of ordinary objects into “smart” entities capable of autonomous communication and interaction.

We will present the various definitions attributed to the Internet of Things as they appear in the literature.

According to the International Telecommunication Union [5], the Internet of Things is defined as a “global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies”. This definition emphasizes the interconnection of objects and the exploitation of their capabilities to offer various services while ensuring security and privacy.

According to Corici et al. [6], the Internet of Things is an evolving field in which hardware and software components model, sense, or influence properties of the physical world. To exploit the potential of an IoT service, security and flexibility are essential requirements.

Sha et al. [7] described the Internet of Things as an emerging paradigm, considering it as the third major wave of innovation in the field of information technology, succeeding the Internet and mobile computing technology. Reference [8] presented compelling data confirming strong growth in the IoT market. Their analysis revealed an exponential increase, forecasting a value reaching several billion US dollars. In addition, the study detailed the progress of IoT in various economic sectors. Significant growth in the automotive, healthcare, and lifestyle fields highlights the widespread expansion of IoT into practical applications that influence our daily lives. Furthermore, Gormus et al. [9] added another dimension to our understanding by explicitly showing the diverse sectors covered by the IoT. This broad coverage confirms that the IoT is everywhere, transforming many sectors of the economy and society. In healthcare, the IoT has introduced remote monitoring systems and connected medical devices that enable continuous monitoring and optimization of care decisions. These advances, supported by the work of Ibrahim et al. [10], promote proactive health management with rapid, secure access to medical records. In the urban planning sector, the impact of IoT is embodied by the development of smart cities. The integration of a network of sensors and actuators, as demonstrated in the study by Pérez and Rodríguez [10], enables efficient management of urban resources, optimizing environmental parameters such as temperature and humidity. This improved management translates into a higher quality of life for city dwellers. In addition, the IoT has also proved its usefulness in the creation of Smart Homes adapted to people with disabilities, a concept put forward by Hussein et al. [11]. These smart homes, designed to meet the specific needs of individuals, illustrate the personalized and humanized application of the IoT. This information, consolidated by IDC [8] forecasts of 41.6 billion connected IoT devices by 2025, underlines the scale of the phenomenon. As the cornerstone of modern societies, the IoT is at the heart of the synergy between technology and infrastructure, propelling our societies into an era of increased connection and intelligence. The rapid expansion of IoT technology can be explained by several factors. Perera et al. [2] identified the evolution of wireless technologies, increased Internet bandwidth, and reduced technology costs as key drivers of IoT growth. In addition, Atzori et al. [12] highlighted the importance of interoperability in IoT, enabling heterogeneous devices to collaborate and paving the way for innovative applications and enhanced services.

3.2. Basic Architecture of IoT

The architecture of the Internet of Things (IoT) is often described in terms of layers to simplify its understanding and development. Although there is no single architecture standard, and developing one is complex due to the natural fragmentation of potential applications, we can conceptualize it in three main layers: the perception layer, the network layer, and the application layer [13].

The perception layer is responsible for collecting physical information using sensors, RFID, GPS, and other devices, converting it into digital data for processing and analysis [14].

The network layer handles data transmission and processing, efficiently transmitting data collected by the perception layer to information processing devices, including cloud servers. It uses various communication protocols such as Wi-Fi, Bluetooth, Zigbee, LoRa, and others [15].

The application layer delivers practical, interactive services and solutions in fields such as healthcare, home automation, and energy management, making IoT tangible for end-users.

This three-layer structure Figure 4 simplifies the complexity of IoT systems and enables a better understanding of their operation and management. It also highlights the importance of harmonious integration between layers to ensure optimal efficiency and security of IoT systems.

In our study, we will use this basic architecture of IoT systems as the basis for our analysis.

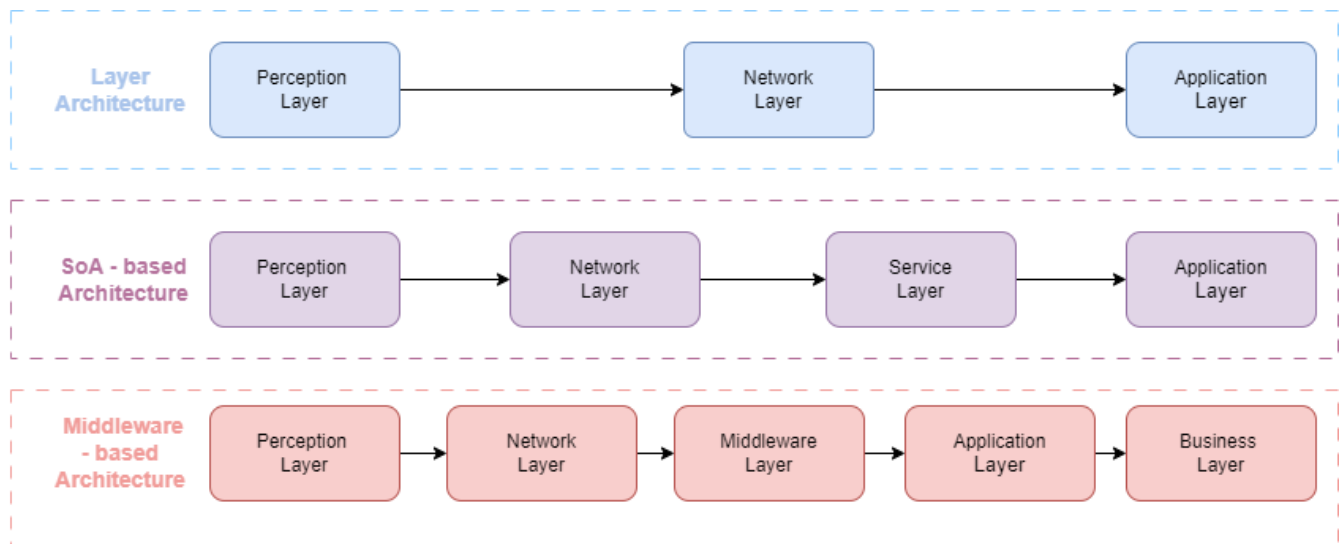


Figure 4. Common IoT architectures.

4. Classification of IoT Attacks Based on Vulnerabilities and Layers

In this section, we present a comprehensive classification of IoT attacks by examining both the vulnerabilities they exploit and the specific layers of the IoT architecture they target. This dual classification is essential for comprehensively understanding potential threats and developing more effective defense strategies.

4.1. Categorization of Vulnerabilities

IoT systems exhibit several vulnerabilities, which can be categorized based on the different layers of the IoT architecture:

4.1.1. Physical Layer

The security of connected objects presents several significant challenges, such as resource constraints and data storage vulnerabilities [6,16–18]. Jing et al. [18] highlighted the physical vulnerability of IoT devices, noting that direct access to devices can lead to serious security breaches, especially when placed in public or unmonitored areas [19].

Hardware limitations make it difficult to implement effective security measures, and vulnerabilities in securing locally stored data pose significant risks [12,15].

4.1.2. Network Layer

The network layer faces vulnerabilities related to insecure communication protocols, diverse connectivity standards, and weak authentication mechanisms [11,16]. Insecure default configurations and insufficient network segmentation further increase the risk of unauthorized access [15].

4.1.3. Application Layer

Application layer vulnerabilities include a lack of updates, third-party application risks, and weaknesses in cryptographic implementations [12,16]. Neglect in updating security patches and the use of weak passwords facilitate unauthorized access, highlighting the need for robust authentication and authorization systems [15].

We introduce a taxonomy Figure 5 that allows us to visualize these vulnerabilities by layer.

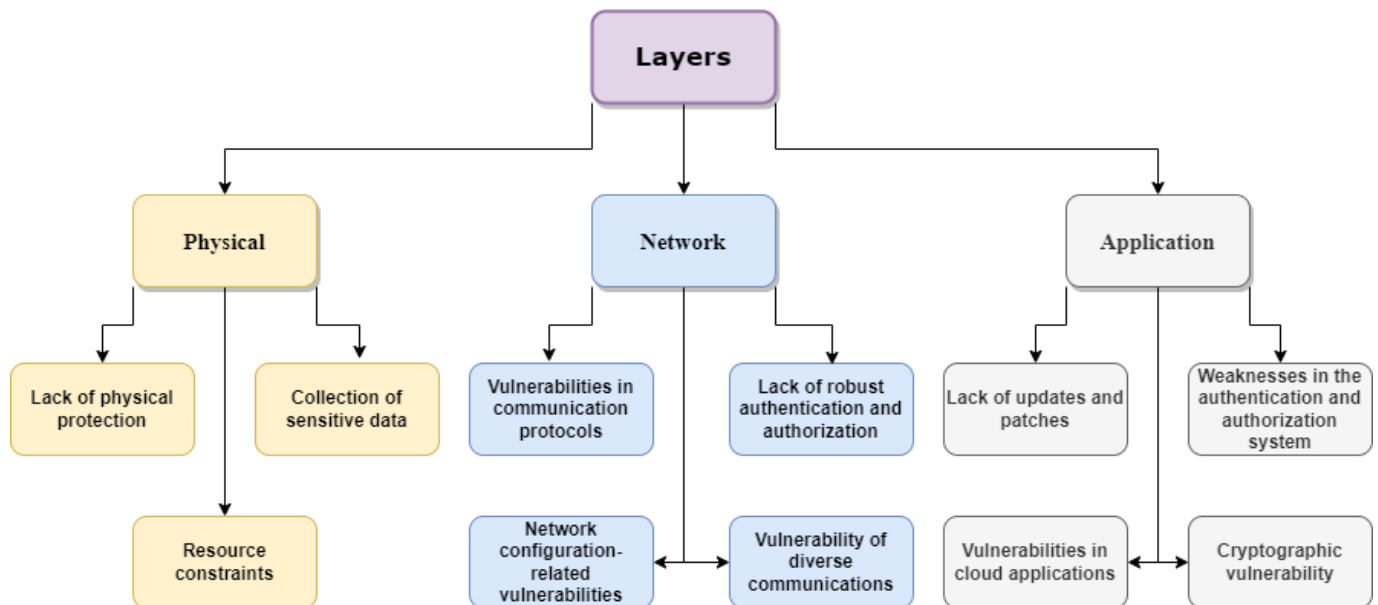


Figure 5. Classification of vulnerabilities by layer.

4.2. Categorization of Attacks

There are multiple attacks on IoT and IIoT objects, which can be classified according to vulnerabilities or according to layers.

4.2.1. Perception Layer

Attacks at the perception layer, such as sensor tampering and side-channel attacks, compromise the integrity of collected data and the functioning of IoT devices [20–22]. Physical attacks like destruction or theft further threaten device security [23].

4.2.2. Network Layer

The network layer is vulnerable to DoS, DDoS, and identity spoofing attacks, which disrupt communication and access to services [14,21,22,24]. Man-in-the-middle and sniffing attacks also compromise the confidentiality and integrity of data exchanges [22].

4.2.3. Application Layer

Application layer attacks include code injection, ransomware, and data theft, targeting the security and availability of IoT applications [16]. Buffer overflows and social engineering further highlight the need for secure development practices and user awareness [12].

Below, we present a Figure 6 summarizing all the layer attacks.

Following the previous classification of attacks by layers in IoT systems, we have developed a detailed taxonomy Figure 7 that organizes these attacks based on the specific vulnerabilities they exploit. This structuring allows us to clearly illustrate the direct links between vulnerabilities inherent in IoT technologies and the various types of attacks identified.

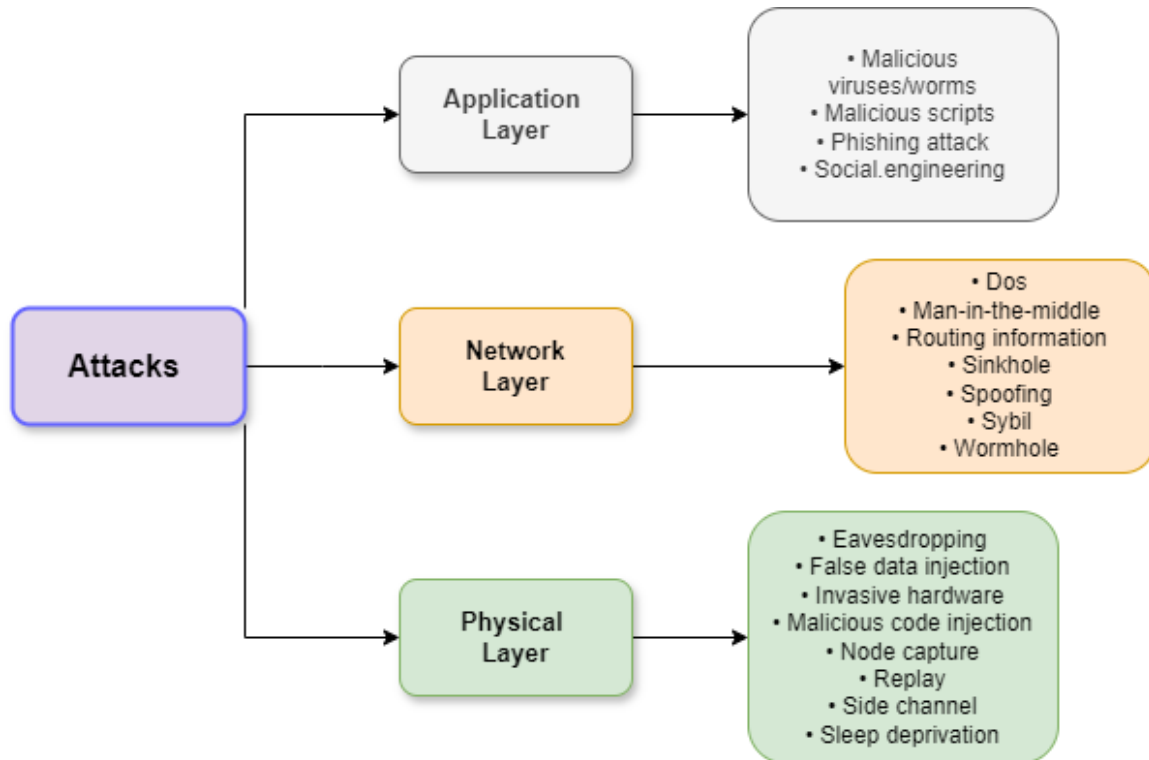


Figure 6. Classification of attacks by layers.

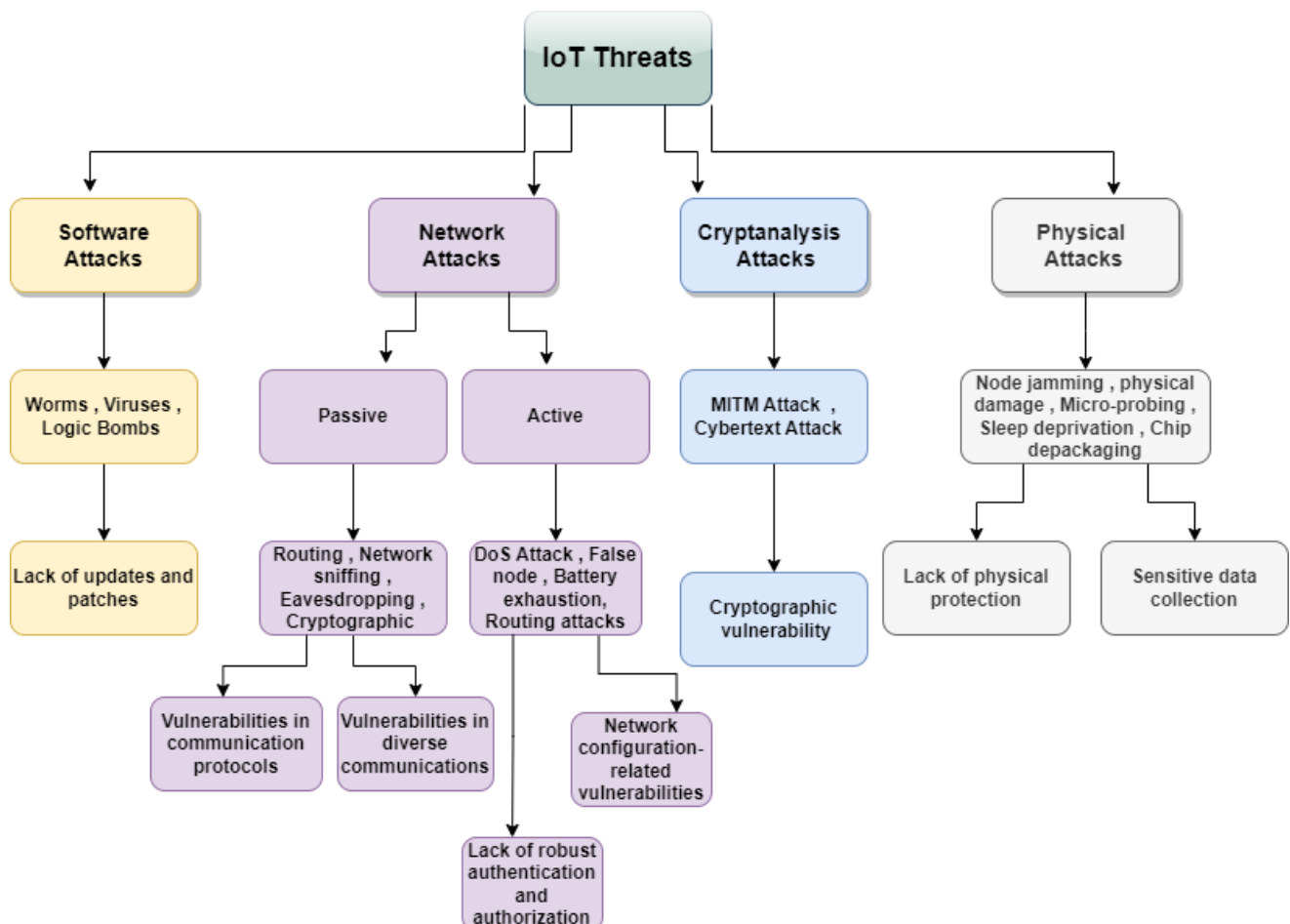


Figure 7. Taxonomy of attack classification according to vulnerabilities.

5. Traditional Intrusion Detection Methods

Intrusion detection systems (IDSs) are essential tools for identifying potential threats and ongoing attacks in IoT systems. Traditional detection methods can be divided into two main categories based on the source of the analyzed data and the detection method.

5.1. Classification Based on Data Source

Among the methods based on data sources, IDSs can be divided into host-based (HIDS) and network-based (NIDS) methods.

NIDSs (network intrusion detection systems), as described by Liu and Lang [25], monitor overall network traffic to detect malicious or suspicious activities. They analyze packets circulating through the network for malicious behaviors or anomalies. These NIDSs can detect attacks occurring between IoT devices and network nodes, and they are effective in identifying suspicious activities that may go unnoticed at the individual host level. The majority of network-based IDSs are independent of the operating system; thus, they can be applied in different operating system environments. The disadvantage is that they only monitor traffic passing through a specific network segment.

HIDSs (host intrusion detection systems), on the other hand, are host-based systems installed on individual IoT or IIoT devices. They monitor host-specific activities and behaviors, such as file changes, unauthorized access attempts, or abnormal operating system behaviors. HIDSs can detect attacks targeting a specific IoT device or resource, providing additional protection at the local level [25]. The drawbacks are that host-based IDSs consume host resources, depend on host reliability, and are unable to detect network attacks.

Hybrid deployment-based detection systems combine both network-based (NIDS) and host-based (HIDS) deployment elements. This approach allows for the benefits of both methods and enhances the overall system security. The Table 3 summarizes the difference between NIDS and HIDS.

Table 3. Comparison between NIDS and HIDS.

Criteria	NIDS	HIDS
Surveillance type	Global network traffic surveillance	Surveillance of specific host activities
Data source	Network traffic	Operating system or application program logs
Detection scope	Malicious or suspicious activities in network traffic	File modifications, unauthorized access attempts, abnormal system behaviors [25]
Operating system independence	Independent of the host operating system	Dependent on the host operating system
Detection target	Attacks between IoT devices and network nodes	Attacks specifically targeting an IoT device or resource [25]
Detection efficiency	High, can detect real-time attacks	Low, needs to process numerous logs [25]
Intrusion traceability	Traces intrusion position and time-based on IP addresses and timestamps	Traces intrusion process based on system call paths
Limitation	Monitors only traffic passing through a specific network segment	Cannot analyze network behaviors

5.2. Classification Based on Detection Method

Several intrusion detection methods are used to detect intrusions in the IoT and IIoT domains.

Anomaly-based method.

The first method is based on behavior modeling, detecting anomalies that correspond to abnormal behavior. This method uses statistical models to establish a baseline of normal behavior for IoT devices. By continuously monitoring data flows, traffic patterns, and communication schemes, the system can detect anomalies that may indicate suspicious or malicious activity. For example, if an IoT device starts generating abnormally high traffic volume or communicating with unusual destinations, this could indicate an intrusion attempt. According to Liu and Lang [25], anomaly detection is particularly useful for identifying unknown attacks and new or emerging behaviors.

Misuse-based or signature-based method.

Misuse detection, also known as signature-based detection, relies on representing attack behaviors as signatures. The detection process involves comparing the signatures of samples with a signature database. The main challenge in building misuse detection systems is designing effective signatures. The advantages of misuse detection lie in its low false alarm rate and its ability to provide detailed information about attack types and their possible reasons. As highlighted by Liu and Lang [25], it has disadvantages such as a high rate of missed alarms, the inability to detect unknown attacks, and the need to maintain a large signature database. In contrast, anomaly detection is based on establishing a profile of normal behavior and then identifying abnormal behaviors based on their deviation from this profile. Thus, the key to designing an anomaly detection system lies in clearly defining a normal profile. The advantages of anomaly detection are its high generalization ability and its ability to recognize unknown attacks. However, it has disadvantages such as a high rate of false alarms and the inability to provide possible reasons for an anomaly. Here Table 4 is a comparison table between anomaly-based and signature-based intrusion detection methods:

Table 4. Comparison between anomaly-based and signature-based methods.

Criteria	Anomaly-Based Method	Signature-Based Method (Misuse Detection)
Operating principle	Modeling normal behavior and detecting deviations	Representing attack behaviors as signatures
Detection approach	Monitoring data flows, traffic models, and communication patterns	Comparing samples with a signature database
Effectiveness against unknown attacks	High	Low
False positive management	High false alarm rate	Low false alarm rate
Attack information	Unable to provide precise reasons for detected anomalies	Provides detailed information on attack types and possible reasons
Main challenges	Clearly defining a normal behavior profile	Designing effective signatures
Advantages	High generalization capability, recognizes unknown attacks [25]	Low false alarm rate, detailed information on attacks
Disadvantages	High false alarm rate, difficulty in identifying reasons for anomalies	High rate of missed alarms, unable to detect unknown attacks, need to maintain a large signature database

5.3. Limits of Traditional Approaches and Comparison with Machine Learning-Based IDS

Traditional intrusion detection systems (IDSs), designed for standard networks, face significant challenges in diverse IoT networks [3]. Their lack of flexibility toward the variety of IoT devices and protocols reduces their effectiveness. Advanced attacks, including zero-day attacks, easily surpass these IDS, which rely on known signatures. Additionally, their inability to handle large volumes of data leads to performance and scalability issues. Their dependency on manual signature updates increases the vulnerability of IoT systems. Traditional IDSs also struggle to differentiate between legitimate and malicious behaviors in varied IoT traffic patterns, increasing the risk of errors. Privacy concerns and integration challenges with IoT further complicate their use. These limitations indicate the need for approaches more suited to IoT.

In contrast, machine learning-based IDSs offer more flexible, scalable, and intelligent solutions to combat security threats in IoT. The Table 5 compares the two approaches:

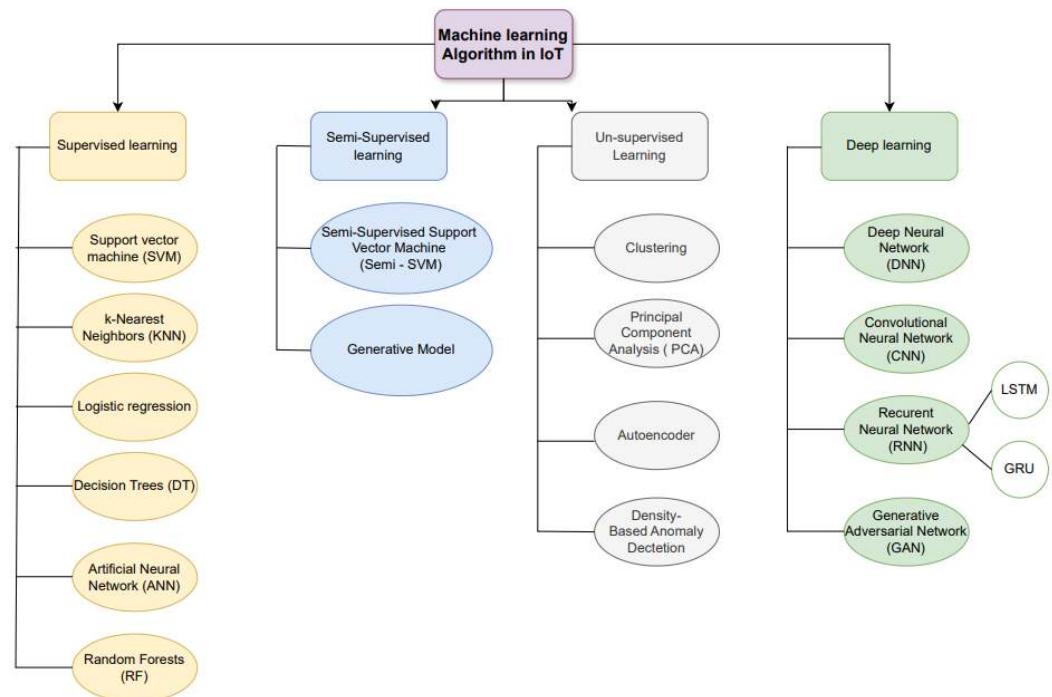
Machine learning-based IDSs can adapt to new behaviors and traffic patterns, making them more suitable for the dynamic and diverse nature of IoT environments. While they may have a higher false positive rate, this can be managed with careful tuning and continuous learning. By integrating these more advanced techniques, IoT systems can achieve a higher level of security, effectively addressing the limitations of traditional IDS.

Table 5. Comparison between traditional and machine learning-based IDSs.

Criteria	Traditional IDS	Machine Learning-Based IDS
Flexibility	Limited, depends on known signatures	High, can detect unknown behaviors
Scalability	Limited, performance issues with large data volumes	Good, handles large data with appropriate resources
Dependency on updates	High, requires manual signature updates	Low, learns continuously from new data
Detection of unknown attacks	Low, does not detect zero-day attacks	High, detects anomalies and new attacks
False positive rate	Low for known attacks, high for new ones	Variable, high for anomalies but manageable
Attack information	Detailed for known attacks	Limited but can be improved with interpretability techniques

6. Machine Learning for Intrusion Detection

In the field of intrusion detection, particularly in IoT systems, machine learning has established itself as a leading technology. This branch of artificial intelligence enables systems to automatically detect suspicious or malicious activity by learning from available data [26]. There are several approaches to machine learning for intrusion detection, each with its own strengths and limitations depending on the application context. Figure 8 presents the taxonomy of different machine learning methods used in IoT.

**Figure 8.** Taxonomy of machine learning methods in IoT.

7. Supervised Learning

Supervised learning relies on the use of labeled data, where each data point is associated with a label indicating whether it represents an intrusion or normal activity. The model learns to make this distinction by analyzing the features of the training data and is then able to predict labels for new data.

7.1. Popular Methods and Applications

7.1.1. Artificial Neural Networks (ANNs)

ANNs are trained to recognize specific attack signatures and abnormal behavior by analyzing network traffic [27]. According to research by Sohaib et al. [28], this method has demonstrated an average accuracy of 84% and a low false-positive rate, at less than 8%,

in repeated cross-validation for intrusion detection in IoT systems. Arul Anitha et al. [27] also proposed an artificial neural network-based intrusion detection system using a multi-layer perceptron (MLP) approach to detect DOS and version attacks in IoT. They captured 73,880 data packets in their simulation, of which 1307 were malicious. The MLP model trained correctly classified malicious and normal packets, with a very low error rate. L. Jamal et al. [29] also used ANNs to detect malicious behavior on an IoT network dataset consisting of 461,043 records in total, of which 300,000 were benign and 161,043 were malicious. With the proposed methodology, malware was detected with an accuracy of 94.17% and classified with an accuracy of 97.08%. These results underline the ability of ANNs to efficiently process large heterogeneous datasets. However, a major challenge remains regarding the accurate distinction between malicious and normal data packets, especially in the case of large volumes of data processed by the ANN. The authors highlighted the effectiveness of their model in detecting intrusions with high accuracy while noting the need for more research to apply it to real data [27].

7.1.2. Support Vector Machine (SVM)

SVMs are widely used for intrusion detection in IoT systems. Goeschel et al. [30] demonstrated the effectiveness of SVMs, achieving an accuracy of 99.62% with a low false positive rate of 1.57%. Jing et al. [18] used SVMs for intrusion detection with the UNSW-NB15 dataset, implementing a nonlinear scaling method that improved accuracy to 85.99% for binary classification and 75.77% for multiclass classification. SVMs have also been applied to secure intelligent networks, efficiently detecting known and unknown attacks and outperforming traditional methods. Ioannou et al. [31] evaluated two SVM approaches (C-SVM and OC-SVM) for detecting network attacks in IoT systems. C-SVM showed classification accuracy of up to 100% with unknown data from the same network topology and 81% accuracy in an unknown topology, while OC-SVM achieved a maximum accuracy of 58%. These results confirm the potential of SVM as a robust tool for anomaly detection in IoT network security [32]. However, SVMs can be less flexible with highly varied and dynamic IoT data. Building and optimizing SVM models, especially for large and complex datasets, can be computationally expensive. Further research is needed to improve their accuracy in unknown environments.

7.1.3. K-Nearest Neighbors (KNNs)

K-Nearest Neighbor (KNN) is a non-parametric classification method effective for tasks such as intrusion detection in IoT systems [33,34]. Abdaljabar et al. [34] implemented an approach combining KNN and Decision Tree algorithms, achieving an accuracy of 100% after data normalization, proving effective in detecting IoT attacks. Wenchao Li et al. [35] developed a similar system for wireless sensor networks, focusing on accuracy and speed. Govindarajan et al. [36] adopted a hybrid k-NN method, incorporating cross-validation to improve performance, despite the lack of precision on the datasets used. Another study by Aref et al. [37] introduced a semi-supervised approach with hyper-parametric KNN using the NSL-KDD dataset, achieving 95% accuracy. These studies highlight the effectiveness of KNN in intrusion detection but also its limitations, such as sensitivity to the choice of the parameter 'k' and susceptibility to noise in the data.

7.1.4. Logistic Regression (LR)

Linear regression has been effectively used in identifying specific threats within IoT environments. Bapat et al. [38] used LR to identify malicious botnet traffic, demonstrating its usefulness in threat detection. Swathi Sambangi et al. [39] explored the use of multiple linear regression to detect DDoS attacks in cloud environments using the CIC-IDS 2017 dataset. They applied a feature selection technique to identify the most relevant attributes for predicting such attacks. The model achieved an accuracy of 73.79% using the 16 selected attributes, indicating its robustness for detecting DDoS attacks in cloud environments.

7.1.5. Decision Tree

Decision trees are highly effective in classifying and detecting abnormal behavior in IoT environments. Qaddoura et al. [40] used decision trees to examine network data flow, identifying suspicious traffic sources and helping to combat distributed denial-of-service (DDoS) attacks. Ingre et al. [41] developed an intrusion detection system (IDS) using a decision tree for the NSL-KDD dataset. They applied the correlation feature selection (CFS) method to enhance predictive performance, achieving high detection rates and accuracy, particularly in binary classification. Kajal Rai et al. [42] implemented the DT C4.5 algorithm for intrusion detection in IoT systems, focusing on feature selection and optimization. Using the NSL-KDD dataset, they reported satisfactory performance in terms of model building speed, false positive rate, true positive rate, and accuracy. However, the performance was influenced by dataset size and feature number. These studies demonstrate the adaptability and effectiveness of decision trees in complex applications. However, challenges such as difficulty in identifying new attack forms and dependence on proper feature selection can affect their accuracy in complex cybersecurity environments.

8. Unsupervised Learning

Unsupervised learning is a crucial approach in intrusion detection, particularly useful for identifying abnormal behaviors or anomalies in IoT systems. Unlike supervised learning, unsupervised learning does not rely on labeled data. Instead, it analyzes raw data to discover intrinsic patterns or groupings, which is especially beneficial when data labels are not available or are difficult to obtain.

8.1. Popular Methods and Applications

8.1.1. Clustering

Clustering is a fundamental technique in unsupervised machine learning. It involves grouping a set of data in such a way that the data in the same group (called a cluster) are more similar to each other than to data in other groups [43]. Common examples include K-means and hierarchical clustering [43]. In intrusion detection, these methods can identify clusters of similar behavior, enabling the detection of anomalous activity that deviates significantly from the established clusters. Muniyandi et al. [44] proposed an anomaly detection method combining K-means and the DT C4.5 algorithm. However, they found that the performance of K-means was less effective than that of supervised learning methods, particularly for the detection of known attacks. In parallel, Peng et al. [45] proposed an improved K-means-based detection method with a mini-batch for processing large datasets, such as the KDD99 dataset. K-means clustering has also been used to detect intrusions, specifically Sybil attacks in WSNs [35], with interesting results. This study suggests the use of channel vector clustering to distinguish Sybil attackers from normal sensors.

8.1.2. Principal Component Analysis (PCA)

PCA is a dimensionality reduction method that has been applied in IoT for efficient data processing. Peng et al. [45] used PCA on the KDD99 dataset, transforming features into numerical types before clustering them with improved K-means. This approach enhances clustering performance and reduces computational load. Zhao et al. [15] proposed a model using PCA for dimensionality reduction, combined with softmax regression and the KNN algorithm. Integrating PCA with these classifiers resulted in a system that is both efficient and low on computational resource demands, capable of operating in real-time in IoT environments.

However, PCA can lead to a loss of important information and assumes the linear independence of the principal components, which limits its application in some complex contexts.

8.1.3. Autoencoders

In the context of the Internet of Things (IoT), wireless sensor networks (WSNs) in particular, autoencoders have demonstrated their ability to solve complex problems. Luo et al. [46] proposed a model where autoencoders were introduced into WSNs to detect anomalies. This research implemented a two-part solution: anomaly detection on sensors in a distributed mode without the need for communication with other sensors or the cloud, and management of computationally intensive learning tasks on the IoT cloud. This case study illustrates how autoencoders can be effectively applied in WSNs for anomaly detection, highlighting their potential in processing and analyzing complex data generated by IoT devices. Aboelwafa et al. [47] explored the use of autoencoders for the detection of false data injection attacks in the Industrial Internet of Things (IIoT). Their results show that this method outperforms support vector machine (SVM)-based techniques in terms of attack detection and false alarm reduction while demonstrating notable efficiency in corrupted data recovery.

8.1.4. Density-Based Anomaly Detection

Density-based anomaly detection, as with algorithms like K-means and DBSCAN, is effective in grouping data based on salient features and identifying abnormal behaviors. A practical application was carried out by Garg et al. [48], where they used these algorithms to analyze logs and manually determine the specific types of attacks associated with abnormal clusters. In most other studies, this method is combined with other methods to enhance the security of IoT systems.

However, this method can be limited by its dependence on the chosen clustering characteristics and its sensitivity to the algorithm's parameters, such as the neighborhood radius in DBSCAN. Moreover, the need for manual analysis of clusters to identify specific types of attacks can make the process less efficient for large amounts of data.

8.2. Semi-Supervised Learning

Semi-supervised learning is a machine learning approach that combines elements of supervised and unsupervised learning. While supervised learning relies on labeled data for training, and supervised learning works on unlabeled data with an exploratory objective, semi-supervised learning uses both labeled and unlabeled data to train a classifier. This method aims to solve the problem of needing large amounts of labeled data for training in supervised ML, by also incorporating unlabeled data.

However, it is important to note that although semi-supervised learning seems promising in combining the advantages of both approaches, it may not always achieve the detection accuracy offered by supervised learning. Despite this limitation, there have been some successful applications of semi-supervised learning in the field of IoT security [49].

For example, Al-Jarrah et al. [43] developed a semi-supervised multilayer clustering (SMLC) approach for network intrusion detection and prevention. This method has shown its effectiveness in learning from partially labeled data while offering detection performance comparable to supervised ML.

Rathore et al. [21] also proposed a method based on an extreme learning machine (ELM) integrating semi-supervised fuzzy C-means to improve attack detection in IoT.

9. Deep Learning

Deep learning is playing an increasingly significant role in intrusion detection, particularly in complex IoT systems. Deep learning models consisting of neural networks with multiple hidden layers, enable deeper analysis and better feature extraction from data, which is well suited to identifying complex and subtle patterns of malicious activity [50].

9.1. Popular Methods and Applications

9.1.1. Deep Neural Networks (DNNs)

DNNs have demonstrated significant improvements in intrusion detection for IoT architectures by learning directly from raw data. Ahmad et al. [4] showed that using the IoT-Botnet 2020 dataset, DNNs achieved a 0.57–2.6% increase in model accuracy and a 0.23–7.98% reduction in the false alarm rate compared to other methods, confirming their superiority in an IoT network. Jin Kim et al. [51] also studied an intrusion detection system using a DNN on the KDD Cup 99 dataset, revealing high accuracy in intrusion detection with a low false alarm rate. The DNN model achieved a detection rate of 99.95% for abnormal flows, although there was a slight decrease (3.87–10.99%) in the detection of benign flows compared to other algorithms. Despite this, the DNN still achieved a score of 96.085%, outperforming other methods. These results underline the effectiveness of DNNs in intrusion detection while highlighting the need for ongoing optimization to improve benign flow detection. The advantages of this approach include high detection accuracy and adaptability to evolving attacks. However, the complexity of DNNs and the need for large volumes of data for training are seen as disadvantages.

9.1.2. Convolutional Neural Networks (CNNs)

In the IoT sector, CNNs have been used to identify malware on Android systems. Research [33] has revealed that CNNs can automatically learn features relevant to malware detection from raw data, eliminating the need for manual feature manipulation. This technique marks an advance in traditional machine learning methods, offering complete end-to-end modeling. Kim et al. [52] conducted a study on network intrusion detection using a CNN model, focusing on detecting denial-of-service (DoS) attacks by exploiting the KDD CUP 1999 and CSE-CIC-IDS2018 datasets. Their method involves transforming data features into images to train the CNN model. The results indicate that the CNN model outperforms a model based on a recurrent neural network (RNN). Chen et al. [24] developed a network intrusion detection system based on a CNN. They trained their model using both extracted features and raw network traffic, demonstrating superior accuracy compared to models based solely on extracted features. They used standard datasets such as NSL-KDD. The main advantage of their method lies in the ability of CNNs to process raw data directly without the need for extensive pre-processing, thus improving detection. However, the complexity and high computational demands of CNNs are seen as constraints.

9.1.3. Recurrent Neural Networks (RNNs)

RNNs are particularly effective for applications in threat detection, where models are highly dependent on the temporal aspect of data. Shin Park et al. explored an intrusion detection method based on LSTM, demonstrating its superiority in terms of mean square error (MSE) and mean absolute error (MAE) compared to other techniques [53]. Tang et al. [54] proposed an intrusion detection method using a deep recurrent neural network (GRU-RNN) for SDNs, achieving 89% accuracy with six raw features. This model proved effective for real-time detection without significantly impacting network performance. Kim et al. [52] presented an intrusion detection approach using LSTM-RNN on the KDD Cup 1999 dataset, highlighting the effectiveness of LSTM-RNN in terms of detection rate and false alarms. Torres et al. [55] demonstrated the effectiveness of RNNs in analyzing network traffic behavior to identify malicious activity, confirming the crucial role of RNNs in the accurate classification of network traffic. Advances in RNN architectures, such as LSTM and GRU, have dramatically improved the ability of neural networks to handle complex data sequences, particularly in areas such as intrusion detection where accuracy and consideration of temporal dependencies are crucial [56,57].

9.1.4. Generative Adversarial Networks (GANs)

In the realm of IoT, GANs prove to be a promising approach for intrusion detection. Ferdowsi et al. [58] developed a distributed GAN architecture for a fully distributed intrusion detection system (IDS) in the IoT environment, allowing each IoT device (IoTD) to monitor its own data as well as that of its neighbors. This method is particularly beneficial for maintaining data privacy. Eunbi Seo et al. [26] designed an intrusion detection system for vehicular networks using GANs, named GIDS, to identify unknown attacks based solely on normal data. Their trials showed high accuracy in detection, although distinguishing normal component failures from deliberate attacks remains a challenge. Dashun Liao et al. [59] proposed a network intrusion detection method based on GANs, focusing on enhancing attack detection by generating new training data. Their approach demonstrated significant improvement over conventional methods when tested with various datasets. Simulation results on a daily activity recognition dataset revealed that the distributed GAN proposed by Ferdowsi et al. [58] provides up to 20% additional accuracy, a 25% improvement in precision, and a 60% reduction in false positive rate compared to a standalone GAN. These results highlight the superior effectiveness of the distributed GAN solution for precise intrusion detection with less dependence on central units and better preservation of data privacy.

Tables 6 and 7 summarize all the methods discussed in the literature.

Table 6. Machine learning methods for intrusion detection in IoT.

Method	Study	Dataset	Attacks and Vulnerabilities Explored	Results
ANN	[28]	UNSW-15 Dataset	Dos, Probe, U2R, R2L	Average precision of 84%, false positive rate < 8%
	[27]	Simulated with Contiki OS/Cooja Simulator 3.0	DIS attack, Version attack	Accurate classification, low error rate
SVM	[30]	KDD Cup 99	Various types of attacks	Significant reduction in false positives
	[18]	UNSW-NB15	Backdoor, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, Worms	85.99% precision in binary classification, 75.77% in multi-classification
KNN	[36]	DoH20	Various types of attacks	100% precision for KNN and DT
	[34,36]	University of New Mexico data	Dos, Probe, U2R, R2L	Reduced execution time up to 0.01%, decreased error rates up to 0.002%
Naive Bayes	[60]	KDD Cup'99	DoS, Probe, U2R, R2L	Improved false positive rates, cost, and calculation time
Logistic Regression	[38]	Malware Capture Facility Project, Stratosphere IPS data	Traffic from 8 different botnet families	AUC of 0.985, precision of 95%, recall of 96.7%
	[39]	CIC-IDS 2017	DDoS and Bot	73.79% precision with information gain-based feature selection
Decision Tree	[41]	NSL-KDD	DOS and DDOS attacks	73.79% precision in DDoS attack detection
	[42]	NSL-KDD	Various IoT attacks	Improved precision and model construction time

Table 7. Machine learning methods for intrusion detection in IoT.

Method	Study	Dataset	Attacks and Vulnerabilities Explored	Results
K-means	[44]	MIT-DARPA 1999 network traffic data	DDOS attacks, code injection	Improved precision, reduced false positives
K-means and PCA	[45]	KDD Cup 99	Various attacks	-
Autoencoder	[46]	Indoor WSN testbed	Various attacks	High detection accuracy, low false alert rate
	[47]	IIoT industry-specific	False data injection attacks targeting IIoT	Significant improvement in attack detection compared to SVM-based methods
DBSCAN	[48]	Not specified	Various varied attacks	Effective data clustering and identification of abnormal behavior
DNN	[4]	IoT-Botnet 2020	Various types of attacks	High precision, adaptability, detects complex patterns
	[51]	KDD Cup 99	-	-
CNN	[52]	KDD CUP 1999 and CSE-CIC-IDS2018	DoS attacks	Effective for malware detection on Android, superior to RNNs for DoS detection
RNN	[24]	CIC-IDS	Various attacks	-
	[53]	NSL-KDD	Various attacks	-
	[54]	NSL-KDD	Attacks in SDN networks	89% precision with only six raw features
GAN	[55]	Bot-IoT Dataset	Botnet behaviors in network traffic	High attack detection rate with low false alarm rate, challenges with indistinguishable and unbalanced traffic
	[58]	Daily activity recognition dataset collected from 30 subjects using a smartphone	Internal and external attacks, including false data injections	Distributed GAN shows up to 20% higher precision, 25% higher recall, and 60% lower false positive rate compared to standalone GAN
	[59]	KDD Cup 99	-	Excellent results for intrusion detection, with approximately 99% precision for all cases and high detection rate

9.2. Review of Datasets

The evolution of intrusion detection datasets illustrates changes in threats and technological advancements in the IoT domain. Through the literature review, we present the following Table 8, which summarizes the timeline of the datasets, the types of attacks they include, and the studies that used them.

Table 8. Datasets in IoT.

Dataset	Attack	Data Size	Data Type	Study
DARPA1998	Dos, Probe, U2R, R2L	Varies	Raw packets	[44]
KDD Cup 99	Dos, R2L, U2R, Probing	4,730,503 packets	Network records	[45,52,59–62]
NSL-KDD	DoS, R2L, U2R, Probe with 22 types of subcategories of attacks	149,470	Network records	[41,42,53,54]
UNSW-NB15	Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms	2.5 GB	PCAP, CSV	[18,28]
CIDDS-001	Port scan, Dos, Ping of Death, etc.	700 MB	Data flows, CSV	[63–65]
CIC-IDS-2017 and CSE-CIC-IDS2018	Bot, brute force, DoS, Infiltration, SQL injection	Varies greatly	PCAP, CSV	[18,39]

Table 8. Cont.

Dataset	Attack	Data Size	Data Type	Study
BoT-IoT	DDoS, DoS, Reconnaissance, Theft	5 GB	PCAP, CSV	[4,55]
Edge-IIoTset	DDoS_UDP, DDoS_ICMP, SQL_injection, Password, Vulnerability_scanner, DDoS_TCP, DDoS_HTTP, Uploading, Backdoor, Port_Scanning, XSS, Ransomware, MITM, Fingerprinting	Varies also	Network traffic flows, Security event logs, IoT device metrics, Specific attack records, Web traffic data, Communication metadata	No studying

10. Discussion

As we deepen our understanding of intrusion detection systems in IoT environments, it is crucial to examine the challenges and current limitations, as well as the trends and emerging innovations shaping this rapidly evolving field.

10.1. Challenges and Current Limitations

In the field of intrusion detection for IoT systems, several major challenges need to be overcome to improve the effectiveness of detection models. The first challenge concerns the creation and updating of datasets. Currently, the lack of diversity and representativeness in datasets limits the effectiveness of intrusion detection systems (IDSs), as they struggle to reflect the complexity of IoT environments. In addition, the various models proposed in the literature are often tested on heterogeneous data, making it difficult to establish a standard comparison indicator to assess their performance. Given these limitations, it is essential to develop datasets specifically adapted to IoT networks, taking into account their unique characteristics and specific threats. Constant updating and evolution of these datasets are also necessary to keep pace with current trends in cybersecurity attacks, which implies the integration of more sophisticated and diversified attack scenarios. It is also crucial to achieve adequate data balance and representativeness. This means ensuring a balanced distribution between different types of attack and normal traffic, to enhance the reliability of IDSs. Careful selection of features is also very important for optimizing models for intrusion detection and making them less complex. Optimizing hyperparameters in hybrid models is another obstacle. The complexity of these architectures makes their calibration difficult, directly affecting the reliability of IDSs, especially in resource-constrained environments. The generalization of IDS models to real-life scenarios remains a major challenge. Indeed, the performances obtained on laboratory datasets do not necessarily translate into real-world environments. This situation underlines the need to adapt and test models under conditions more representative of the real world. Finally, IDS models in IoT systems face significant resource constraints. The limited computing and storage capacity of IoT devices restricts the implementation of complex and sophisticated models, directly impacting the sophistication and effectiveness of detection methods.

10.2. Current Trends

Current trends in intrusion detection in IoT systems are marked by rapid evolution due to advances in technologies and methods. The adoption of advanced artificial intelligence techniques, such as deep and unsupervised learning, is radically transforming the ability of intrusion detection systems (IDSs) to analyze and understand data. These techniques enable more accurate and rapid identification of anomalies, which are essential for the security of IoT networks. New perspectives arise from the integration of federated learning and reinforcement learning. These innovative techniques contribute to the development of more privacy-friendly and adaptive IDS models, responding to the growing complexity of security threats in IoT environments. Class balancing and feature selection have become unquestionable milestones. In addition, optimizing the hyperparameters of IDS models

is now a priority. Such precision in model tuning is essential for adapting to the specific challenges of IoT environments. Finally, there is a growing trend toward real-time detection and the creation of fast, reactive models. An essential element of proactive security for IoT systems is the acceleration of intrusion detection and response through parallel processing and edge computing. These trends illustrate the crucial importance of continuous innovation and cross-industry collaboration in IoT security. They highlight the need for advanced, tailored solutions to overcome current and future challenges, particularly in terms of data protection and compliance with security standards. In applying machine learning models to IoT environments, several unique challenges and requirements must be addressed. The application of machine learning models in IoT environments presents unique challenges and requirements, particularly concerning resource constraints, real-time processing needs, and the diversity of IoT devices. A comparative analysis of different models highlights their strengths and weaknesses with respect to IoT-specific metrics such as energy efficiency, processing time, and adaptability to various types of IoT attacks. Support vector machines (SVMs) offer high accuracy in detecting intrusions and are robust against overfitting with proper kernel selection, but they come with high computational costs and memory usage, and are sensitive to parameter tuning, making them less effective with dynamic IoT data. Decision trees are easy to interpret, have fast training and prediction times, and are effective in feature selection, but they are prone to overfitting and less effective in handling complex and varied IoT data. K-nearest neighbors (KNNs) are simple to implement and effective in detecting anomalies, yet they have high computational costs for large datasets, are sensitive to the choice of 'k', and are impacted by noisy data. Deep neural networks (DNNs) provide high accuracy, are capable of learning complex patterns from raw data, and are adaptable to various types of attacks, but they require large datasets for training, are computationally intensive, and have the potential for overfitting without proper regularization. Autoencoders are effective in anomaly detection, reduce the dimensionality of data, and can handle unsupervised learning, but they may lose important information during compression and are less effective in highly noisy environments. Generative adversarial networks (GANs) can generate synthetic data for training, are effective in detecting unknown attacks, and preserve data privacy, but they have high computational costs, training instability, and require careful balancing of the generator and discriminator. Recurrent neural networks (RNNs) capture temporal dependencies in data and are effective in sequential data analysis, but they are prone to the vanishing gradient problem, have high computational and memory requirements, and are less effective for non-sequential data. To effectively evaluate the performance of machine learning models in IoT environments, it is important to consider IoT-specific metrics. Energy efficiency is critical due to the limited power resources of IoT devices. The processing time is crucial for real-time applications. Adaptability is key for the model's capability to adapt to various types of IoT attacks and different IoT device characteristics. Scalability measures the model's performance as the number of IoT devices and data volume increases. Finally, accuracy, including precision and recall in detecting true positives and minimizing false positives, ensures the chosen machine learning models can effectively enhance the security of IoT systems while meeting their unique constraints and requirements.

11. Conclusions

This literature review highlights the crucial importance of security in the rapidly expanding domain of the Internet of Things (IoT). Through an in-depth analysis, we have explored the inherent challenges in IoT system security, highlighting the vulnerabilities and types of attacks that threaten the stability and reliability of these technologies.

Our study has emphasized the evolution of intrusion detection systems, focusing on the limitations of traditional methods. In this context, machine learning emerges as a promising solution, capable of adapting and responding to the complex and constantly evolving threats in the IoT environment. The review has examined in detail various machine learning approaches, including supervised, unsupervised, and deep learning, evaluating their effectiveness in intrusion detection.

Nevertheless, despite significant advancements, there remain challenges and opportunities for improvement. The effectiveness of models must be evaluated within real-world scenarios, and their large-scale deployment demands careful attention to practical aspects.

Looking forward, this review encourages continued research in the field of intrusion detection for IoT, emphasizing the development of innovative solutions. It is imperative to continue exploring approaches that not only detect intrusions effectively but are also viable within the dynamic and heterogeneous framework of IoT.

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